Now, a thin fiber can be bent around into complicated shapes without any portion of the fiber's material being distorted by anything close to 1 percent. The key is the thinness of the fiber. You could imagine the *Endurance*'s strength relying on a huge number of thin fibers stretching around the ring, like the strands of a cable that hold up a suspension bridge and can bend as needed when a strong wind blows. But that would make the ring *too* flexible. The ring needs much resistance to being deformed, so it won't deform so severely, when subjected to tidal forces, that the modules crash into each other.

The designers, in my interpretation, worked hard to make the *Endurance* resist deformation but be able to deform without breaking if it encounters tidal forces far stronger than anticipated.

Explosion in Orbit Above Mann's Planet

This design philosophy really pays off when Dr. Mann unwittingly triggers a huge explosion that breaks the *Endurance*'s ring, destroys two of the ring modules, and damages two others (Figure 20.2).